

Automation Take-Home Test

Please build out your test suite to the best of your ability. This exercise is designed to take approximately 2 hours, though you may spend more or less time as you feel is appropriate. Please include the approximate amount of time you spent on the exercise in the README.

Technical Requirements

Please use the following stack:

- Playwright
- TypeScript
- Node.js / npm

Include all project setup and dependencies necessary for us to clone the project and execute the tests. Please include a short README with instructions for installing dependencies and running the suite.

Test Scenarios

Setup

This step does not need to be automated, it's just to create the account you'll use for the test cases in Part 1 and Part 2 below.

To create a new test account:

1. Navigate to [Brainpop.com](https://brainpop.com).
2. Click **Login**.
3. On the login modal, select **Enter Code**.
4. Enter the code **ULONUQFL** and follow the prompts to complete account creation.
5. Check the email address you registered with for a confirmation email from BrainPOP, and confirm the account from that email.

Part 1: Login

Some BrainPOP content and functionality require a user to be logged in. Your primary task is to automate and validate the standard BrainPOP login flow.

Create automated tests that cover the login experience. At a minimum, include coverage for:

1. Successful login with valid credentials.

2. Validation that the user is logged in successfully.
3. Login with an invalid username and/or password and validation of the resulting behavior.
4. Attempting to log in without providing all required credentials.
5. Logging out and validating that the authenticated session has ended.

Feel free to include additional login scenarios or validations that you believe provide useful coverage. Where appropriate, consider how you would structure reusable login functionality so that other tests in a larger automation suite could use it without duplicating the login flow.

Part 2: Logged-in User Flow

After successfully logging in:

1. Navigate to a BrainPOP Topic page (example: [Mountains Topic](#))
2. Select the Quiz Feature on that Topic page.
3. Validate that selecting the Quiz Feature navigates the user to the correct page.

The implementation for selecting and validating a Feature should be reusable, so that the same approach can be used to test any Feature available on a Topic page.

What We're Looking For

We are interested not only in whether the tests pass, but also in how you structure your automation. Please use appropriate Playwright patterns and organize the code in a way that could reasonably be extended as the test suite grows.

We will primarily evaluate:

- Test structure and readability
- Appropriate use of Playwright and TypeScript
- Locator strategy
- Reusability and separation of concerns
- Assertions and validation
- Handling of positive and negative scenarios
- Maintainability and scalability of the approach
- Overall reliability of the tests

You do not need to build a large framework for this exercise. Focus on demonstrating the patterns and decisions you would use when contributing to an existing Playwright automation suite.

Use of AI

Using AI tools (Copilot, ChatGPT, Claude, etc.) during this exercise is fine. However, you should be prepared to explain the reasoning behind your decisions and trade-offs, and the resulting suite should reflect sound testing and automation principles regardless of how it was produced. We'll ask about these choices in the follow-up conversation.